Deven Varu

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Education

University of San Francisco (USF) **Aug 2023 – May 2026**
Master in Computer Science *San Francisco, CA*

Experience

SnapLogic **Sep 2025 – Dec 2025**
Software Engineer Intern *San Francisco, CA*

- Researched and implemented the **Agent-to-Agent (A2A) protocol** within the **SnapLogic platform**, translating a newly released specification into a working system.
- Built the **core A2A client and server logic end to end** using **Java**, integrating protocol concepts into SnapLogic **Snaps and pipelines**.
- Designed and validated **real multi-agent workflows**, where a single A2A client coordinated tasks across **four independent A2A servers** hosted on the platform.

Pistachio **Jun 2024 - Aug 2024**
Software Engineer Intern *San Francisco, CA*

- Led a **three-person team** to replace the manual data entry process of extracting product details from **20–200 page furniture catalogs** with an **automated AI-driven solution**.
- **Streamlined text extraction** by testing multiple LLMs (**Gemini, OpenAI, Claude**) and OCR tools, ultimately standardizing on **OpenAI** for consistent JSON output.
- Expanded the solution using **YOLO** model to detect images, extract them, and link catalog visuals with product data for a more holistic dataset, reducing manual effort from **hours to minutes**.

University of San Francisco **Aug 2023 – Dec 2025**
Research Assistant – Full-Stack & Computer Vision *San Francisco, CA*

- Worked on two separate county projects: a **full-stack application for Oakland County** and a **computer vision research project for Boone County, Illinois**.
- For **Oakland County**, built a **Next.js** application with intake workflows, automated data collection across **18 Google Sheets**, email delivery, and a **Pinecone + OpenAI RAG chatbot** for housing-resource questions.
- For **Boone County, Illinois**, researched and tested **YOLO, LaneNet, and DINOv2** on a **7,650-image dataset**, then developed a **YOLOv8** roadway asset detection pipeline with custom visualization and validation tooling, reaching **92% mAP**.

Projects

InterviewWithAI - AI Coding Interview Platform *[Live]*

- Designed a **multi-agent coding interview platform** where a coding agent assists candidates while a voice agent asks **context-aware follow-up questions** based on their reasoning and code.
- Built the **agent orchestration layer with Mastra** to coordinate interview state across agents, with sandboxed **Python and TypeScript execution through Modal** for safe code runs.
- Built a **multi-agent problem generation pipeline** that turns job descriptions into role-specific coding problems through separate **generation, validation, and refinement** stages.

Technical Skills

AI / LLM Systems: AI Agents, Multi-Agent Systems, Agent Orchestration, Voice Agents, A2A Protocol, Mastra, RAG, LLM Evaluation, Prompt Engineering, OpenAI APIs, Vector Databases, Pinecone

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, YOLO, Pandas, NumPy

Programming Languages: Python, TypeScript/JavaScript, Java, SQL, HTML/CSS

Frameworks / Backend: React.js, Next.js, Node.js, Express, NestJS, FastAPI, Supabase

Cloud / Infrastructure: AWS, GCP, Modal, Docker, Kubernetes, GitHub Actions, Axiom